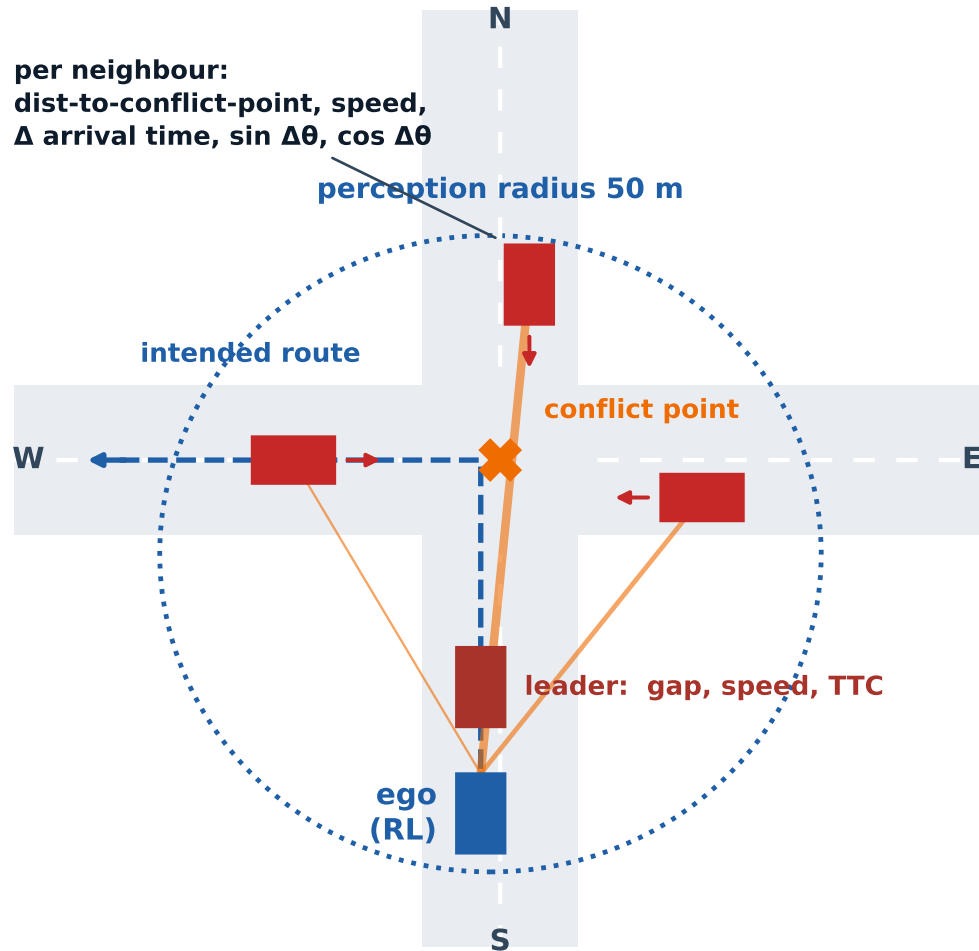
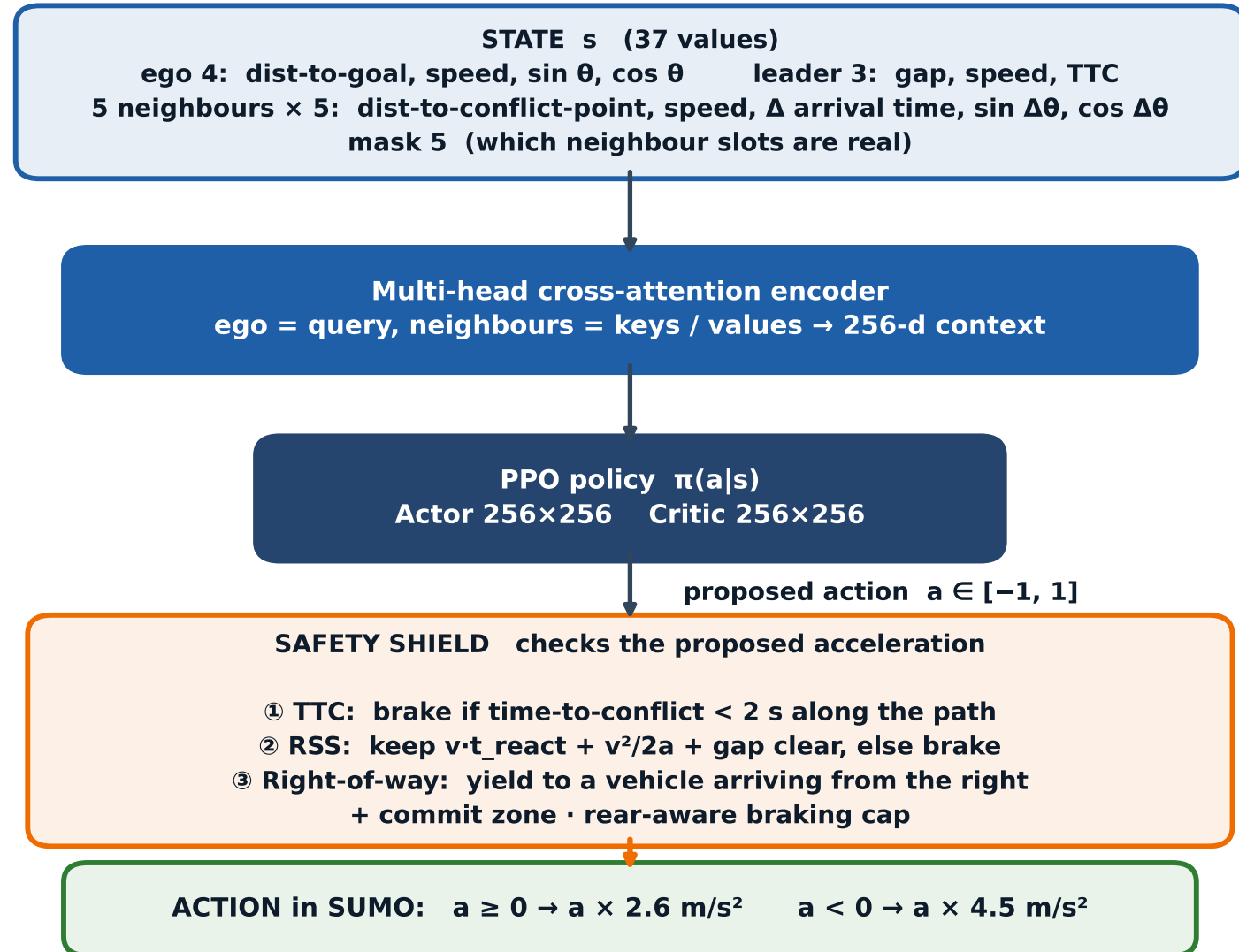


Shielded PPO at an unsignalised junction

The learned policy proposes an acceleration; the shield inspects it and may override it before SUMO applies it



blue = RL ego red = background traffic (IDM)
orange line thickness = attention weight
right-before-left priority, no traffic light



Reward function

Sparse terminal outcome plus dense per-step shaping; the shield adds a small penalty for over-reliance

TERMINAL

collision $r = -10$ goal reached $r = +15$

PER STEP

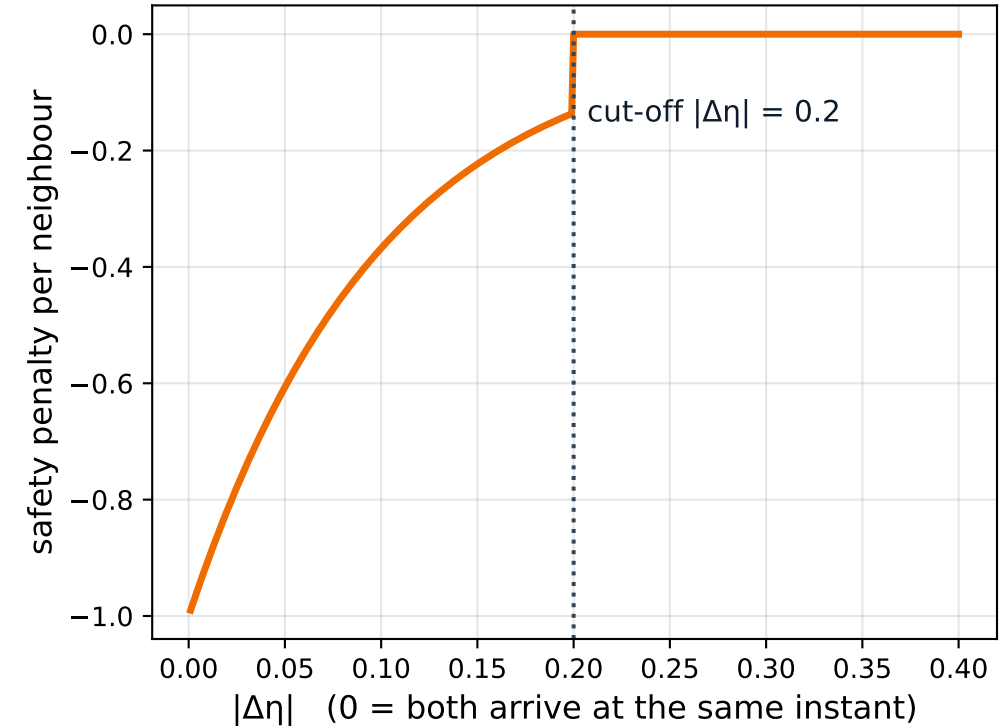
$r = 10 \cdot \Delta\text{progress}$ progress toward the exit
+ $1 \cdot \text{safety_penalty}$ conflict proximity
- 0.01 time penalty

$\text{safety_penalty} = \sum -\exp(-10 \cdot |\Delta\eta|)$ over neighbours with $|\Delta\eta| < 0.2$
- $\exp(-10 \cdot \text{TTC})$ when a leader is within TTC 0.2

$\Delta\eta$ = normalised gap in arrival time at the shared conflict point

shielded variant: extra -0.005 when the shield overrides $> 50\%$ of steps

Penalty spikes as arrival times coincide

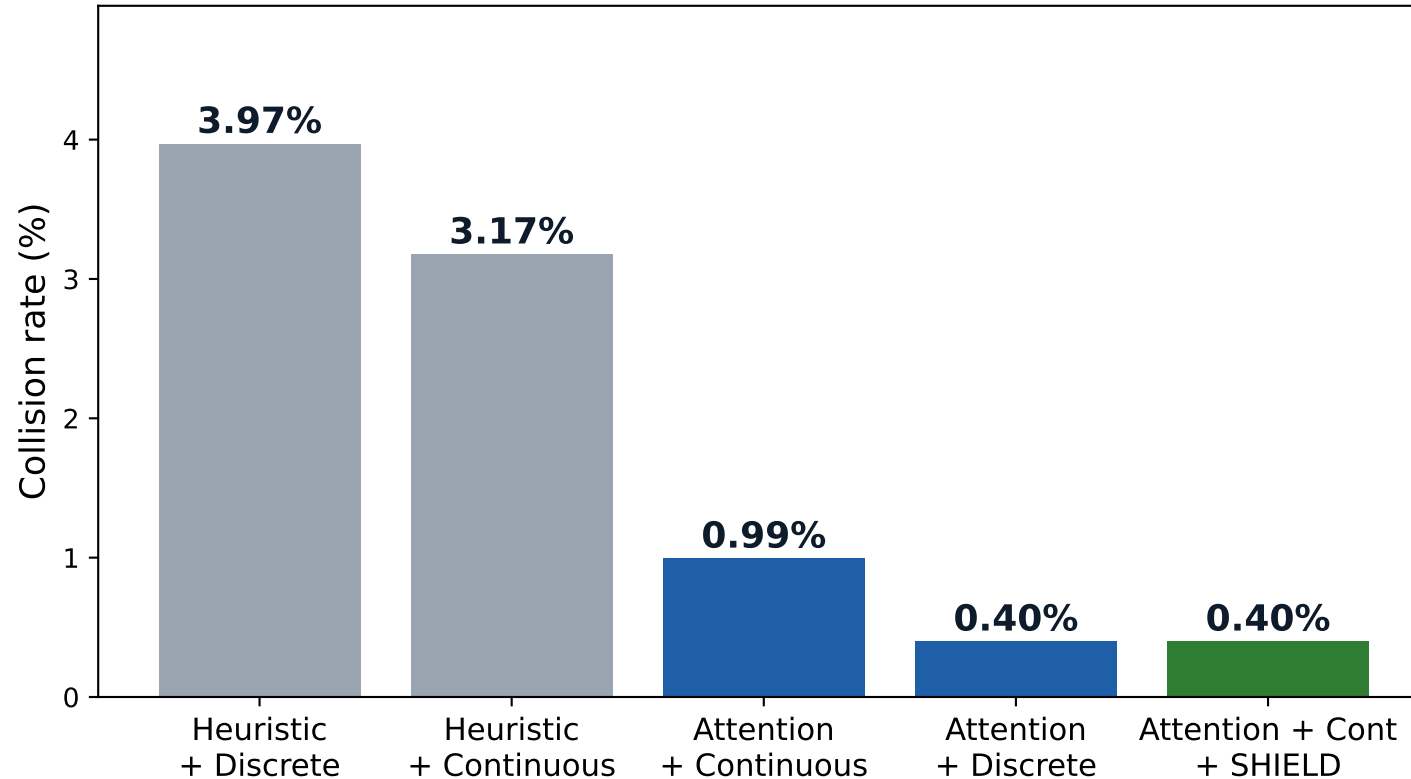


The penalty grows exponentially as two vehicles are projected to reach the conflict point together, so the policy learns to break the tie by slowing or clearing.

Results in our training setup (275 veh/h, rule-abiding traffic)

Evaluated on the pre-defence benchmark: 6 scenarios $\times$ 4 intentions $\times$ 42 episodes = 1,008 episodes per controller

Shield reaches the lowest rate, tied with Attention + Discrete



Why the shield works here

- Shield halves its own baseline:
0.99% $\rightarrow$ 0.40% (10 $\rightarrow$ 4 collisions of 1,008;
 $p = 0.18$, not significant)
- Background vehicles obey SUMO safety rules (speed_mode 31), so they brake for the ego car. Shield braking is therefore safe: it removes crossing conflicts without creating rear-end conflicts.
- Cost: travel time 12.9 s $\rightarrow$ 16.3 s, success 99.0% $\rightarrow$ 97.0%, 18.5% of actions overridden
- Pre-defence report, same benchmark:
best controller 0.88%; heuristic baselines 5.31% and 6.73%

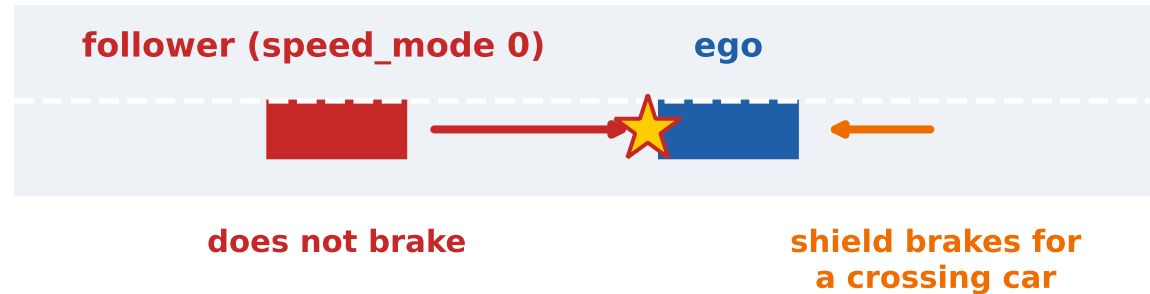
Honest reading: the three attention-based controllers (0.40%, 0.40%, 0.99%) are statistically indistinguishable (all pairwise $p \geq 0.17$). The shield is in the best tier and ties Attention + Discrete; it is not proven better than either attention controller.

Why the shield cannot win in the pre-defence scenario

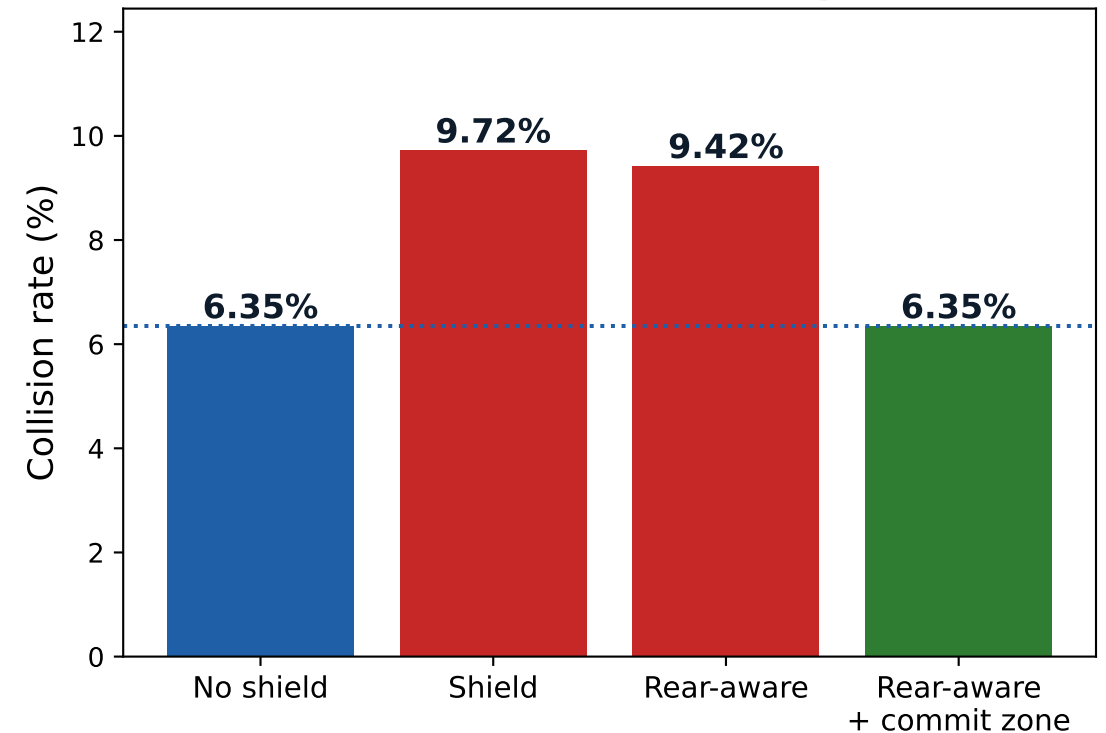
400 veh/h cross traffic, traffic in the agent's own lane, background vehicles that ignore safety rules (speed_mode 0)

The shield's only tool is braking

Slowing down in front of traffic that never yields converts a crossing conflict into a rear-end collision.



Pre-defence environment (1,008 episodes each)



- Original shield made it significantly worse: 6.35% → 9.72% ($p = 0.007$). Left turns, where it brakes most, went 11.1% → 23.8%.
 - Capping the braking (rear-aware) removed some rear-end crashes but let crossing conflicts through: 9.42%, still worse ($p = 0.013$).
 - Adding the commit zone — never brake once the car cannot stop before the conflict point — removed the harm entirely: 6.35%, identical to no shield, with overrides down from 8.8% to 2.9% and travel time from 26.7 s to 23.3 s.
 - Training the policy with the shield active (824k steps, half the budget) reached 12.70%: inconclusive, not a fix.
- Conclusion: the shield helps only where surrounding traffic cooperates. In non-cooperative traffic the best refined shield matches the unshielded controller rather than beating it; a net gain needs the policy to be trained with the shield from the start.